Title Page

"Profile of cervical endurance and peak isometric strength in elite professional male rugby players"

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Profile of cervical endurance and peak isometric strength in elite professional male rugby players

Objectives: To (i) provide position-specific normative data for isometric cervical muscle strength and endurance in professional, male rugby players and (ii) assess the relationship between age, height, weight, and playing position with cervical muscle strength and endurance.

Design: Cross-sectional study

Methods: Professional rugby players completed peak isometric cervical strength testing followed by a test of cervical muscle endurance. Descriptive statistics for continuous variables with strength normalized to bodyweight were performed**.** Multiple linear regression was used to estimate associations between strength measurements.

Results: In total, 136 players participated including front row forwards (27%), other forwards (28%) and backs (45%). Front row forwards had significantly greater peak isometric cervical muscle strength than other position groups, with backs having the lowest strength. Extension produced the highest force for all three position groups (429 N ± 104 N) while flexion produced the least (275 N ± 65 N). Age was associated with increased isometric cervical muscle strength. There was a statistically significant relationship between peak flexion strength and flexion endurance (p=0.003). The average time for the endurance tests were 55.7 (± 17.1) seconds and 52.9(± 20.1) seconds for extension and flexion respectively. Other forwards had lower cervical extension muscle endurance than backs and front row forwards.

Conclusions: Normative values for peak and endurance isometric strength in professional rugby players illustrate significant differences between playing position. Consider age, body weight and intraindividual variability when interpreting cervical strength and endurance results.

KEY WORDS

Rugby, Concussion, Cervical Strength, Neck, Physical testing

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Introduction

Rugby Union (hereafter, rugby) involves frequent exposure to collisions and contact events, resulting in concerns over injury risk, particularly around the head and neck.[1] Despite the success of injury reduction strategies and rule changes in reducing the rate of scrum-related head and

neck injuries,[38] the head and neck still account for approximately 1 in 4 injuries in professional rugby.[36] Fewer injuries occur during set-pieces, such as the scrum, and more injuries are sustained during open play.[46] Concussions most often occur during the tackle event, accounting for 39% of all injuries to the tackler and 17% to the ball carrier.[46]

Cervical muscle strength is a potential risk factor for head and neck injuries in rugby. [10,14,23] Muscle strength and endurance could mitigate the impact of external forces to the head and neck that lead to concussion.[31] Over the past decade, evidence to support increased neck strength and endurance to reduce concussion injury risk has grown.[4,8,10,14,17,19,42,44] There is little consensus around how much neck strength and endurance is required to protect the player, making it challenging to identify players who are at an increased risk of injury or those who have not returned to pre-injury function. Without normative cervical muscle strength and endurance data that are specific to sporting populations, or measurements of pre-injury function, it is difficult to interpret a post- injury assessment. Normative data are available, yet there continues to be a need to include more results from different populations.

Upper and lower limb isometric strength and endurance tests routinely form part of pre-season screening protocols to identify potential injury risk factors.[18] Similarly, a growing body of evidence now supports including cervical muscle strength and endurance tests. [10,22,29]Previous studies measured cervical muscle strength using different devices in different settings,[30,40] while investigating various test positions, including flexed trunk replicating the tackle,[41] seated[1,28] and prone.[5,22] Cervical endurance has also been investigated in numerous ways, usually compared to isometric contraction to failure from a one repetition maximum test.[5,19,22] Recently, guidelines for cervical strength testing were developed progressing our understanding of best practice in this area.[32]

Rugby players are identified as either 'forwards' (positions 1-8) or 'backs' (positions 9-15) due to the physical differences and positional demands observed.[39] However, many studies investigating isometric cervical strength report combined data, which limits interpretation.[21,39] In studies that separated data according to player position, there is little consensus on positional categories, with some studies dividing players into 'forwards' and 'backs'[29] and other into 'front row', 'second row', 'back row' and 'backs'.[30] Therefore, we aimed to provide position-specific normative data for isometric cervical strength and cervical muscle endurance in professional male rugby players using a standardised testing protocol.

2. Methods

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This cross-sectional study assessed cervical isometric strength and endurance in male, professional rugby players with the Irish Rugby Football Union (which includes players from the Republic of Ireland and Northern Ireland). Players contracted to senior (typically contracted players competing at an elite level) or academy squads (younger players who compete for contracts at the elite level) were recruited during the 2017/18 pre-season training period (July and August 2017). Players were tested prior to the start of the rugby season and included in the study provided they were not currently receiving treatment for a cervical spine injury and were able to fully participate in match and/or training activities. Any player who underwent cervical spine surgery in the preceding three months was excluded from the study. All participants provided written informed consent and ethical approval was obtained from the Institutional Review Board, University College Cork Ethics Committee.

Power calculation

We estimated the mean of a continuous outcome variable in a single population, using the formula for determining sample size is given below:

$$n = \left(\frac{Z\sigma}{E}\right)^2$$

where Z is the value from the standard normal distribution reflecting the confidence level that will be used (e.g., Z = 1.96 for 95%), σ is the standard deviation of the outcome variable and E is the desired margin of error.

Testing procedure

Demographic data were collected from all included players, including age (date of birth), height, and weight. The height and weight of players were measured using a calibrated scale, and standard tape measure. Players completed a standardised four-minute warm-up prior to testing, including cervical range of motion into flexion, extension, rotation, and self-resisted isometric holds in each direction.

Participants completed two tests; a peak isometric force test (using the Vald head-harness device described below) and an endurance test adapted from a test used by Halvorson[14] and Peolsson[22] and described in detail below.

Isometric endurance test

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As the endurance test measures the point at which the player fatigues, it was performed after the isometric peak force test.[34] There was a five-minute recovery period between extension and flexion tests, with the extension test carried out first.

The endurance test used a heavy-duty head harness (VodiSport, Ireland) to hold weights on participants' heads in supine and then in prone. A pilot study of 19 players was completed prior to the main study to determine the weight that would ensure the test was both feasible and challenging in this cohort. The pilot group were a representative sample of the final study cohort. A previous study[34] used a 4kg weight for extension and no weight for flexion, in an untrained, non-athletic population. As participants in the current study were highly trained athletes, we hypothesised that 4kg would not prove an appropriate challenge. The aim of the pilot study was to determine the percentage body weight for both flexion and extension tests to produce an endurance test time of 180-240 seconds and 60-120 seconds respectively. We determined that 20% and 7% of the player's body weight, for extension and flexion respectively, was an appropriate weight to challenge all players, without the test taking so long that players lost interest, or being too heavy so that the neck muscles were unable to sustain the weight rather than being fatigued.

The lead researcher spotted the weight throughout the endurance test, while an assistant recorded the time. The same assistant recorded the time using a stopwatch in each case.[15] For extension, the player lay prone with arms by their side, with shoulders and chest on the plinth. The harness was fitted with the player in sitting and the load was applied after the player was lying down (FIGURE 1). Players were instructed to extend their head so that the tip of their chin was pointing at the floor. Players were immediately prompted to reposition if there was any deviation. The test was terminated if the player could not maintain the standard position following two prompts. There was a similar test procedure for flexion, with the player in supine and instructed to flex the upper cervical spine and to maintain retracted cranio-cervical flexion until the player could not maintain the standard position following two prompts (FIGURE 2.). Players were advised to stop when they felt they could not maintain the correct position, when they felt they had reached fatigue or immediately if they felt any pain or discomfort.

FIGURE 1. *Extension endurance test with head harness***.**

FIGURE 2. *Flexion endurance test with head harness*

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Peak isometric strength test

An isometric "make test" was used to measure peak force. Participants generated as much force as possible against a measuring device with no external counterforce. Isometric make tests are more reliable than break tests where participants push against a counterforce until the force exceeds their strength and the contraction "breaks".[6,33,43] While break tests may give higher isometric force scores, we felt a make test was most appropriate for our study.

A reliable testing protocol was used, with good to excellent test-retest, intra and inter-rater reliability comparable to fixed dynamometry (ICC test-retest reliability (0.86-0.92).[20] A single-axis load cell strain gauge (Vald Performance Pty Ltd., Australia) measured maximal voluntary isometric force with the player in an upright seated position with the head in neutral position, hands by the side holding onto the seat base, feet hip width apart and flat on the floor throughout the test.[20] Participants were tested in four directions (flexion, extension, right and left lateral flexion) and the order of testing was pseudo-randomised in each case. A slack harness was placed around their forehead across the frontal and parietal bones above the ear line. The strain gauge was in line with the direction of movement. Participants were given a verbal three second countdown and then asked to pull, producing maximum possible force for four seconds and then told to rest. Thirty seconds rest was given between each of three repetitions, with only the highest score counted.

Data analysis

Data were analysed using SPSS (IBM SPSS Statistics for Windows, Version 24.0. Armonk, NY). Players were analysed according to positions: front-row forwards (1-3), other forwards (4-8) and backs (9-15). The forwards were split into two groups due to the differing game-demands on these players, for example the specific role of the front row forward in the scrum[47] and the fact that other forwards have the greatest workload and highest tackle rate.[26]

Continuous variables were described using means, standard deviations (SD), and full ranges. Comparisons were performed for absolute strength and normalized to bodyweight by dividing strength and endurance results by body weight. Categorical variables were described by counts and percentages in each category. Differences in key variables across position groups were tested using analysis of variance (ANOVA) for continuous variables and Chi-squared tests for categorical variables. A p-value of less than 0.05 was considered statistically significant. Effect size, which is the quantitative measure of the strength of the observed occurrence between groups, was calculated using Cohen's ω^2 for the ANOVA table. The effect size was interpreted as small (>0.01), medium (>0.059), or large (>0.138).[11]

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Multiple linear regression was used to estimate associations between strength measurements. In all models, interactions by position group were evaluated by likelihood ratio tests of two models: one including the relevant product interaction term, and one without. For comparisons where the resulting p-value was >0.05, the interaction term was excluded from the final, reported model. Estimated regression coefficients were reported with 95% confidence intervals, and exact p-values.

3. Results

In total, 136 male professional rugby players (mean age 25.3±3.4 years, height 186±7 cm, weight 103±13.2 kg) were recruited (TABLE 1, Supplementary FIGURE 1). The power calculation suggested that a sample of n=96 was sufficient. The isometric endurance test was completed by all participants, and 129 participants also completed the isometric strength tests; seven players did not complete the isometric strength test. The distribution of isometric endurance strength among different playing positions is provided in Supplementary FIGURE 2.

Cervical extension produced the highest force for all three groups while flexion was the lowest (TABLE 2). There were large effect sizes for peak isometric flexion and extension strength, and for right side flexion strength. All other movements and adjusted to bodyweight had small to medium effects (TABLE 2). Backs had the lowest peak isometric strength scores; front row forwards were significantly stronger in all directions ($p<0.05$). Front row forwards and other forwards were significantly stronger for both flexion and extension peak isometric strength, as well as lateral movements, compared to the backs ($p<0.05$, TABLE 2). For each position group, the average individual scores for right and left side flexion were within 15N. There were significant differences in lateral flexion scores between all position groups ($p<0.001$). The distribution of peak isometric strength in all four directions per playing position are in Supplementary FIGURE 3.

Mean cervical muscle endurance test scores were 55.7 (± 17.1) seconds for extension and 52.9 (± 20.1) seconds for flexion. While other forwards had less extension endurance compared to backs and front row forwards, this was not statistically significant (TABLE 2).

The distribution of strength across both tests is presented through density plots in Supplementary FIGURE 4, with greater peak isometric strength for front row, while lesser peak isometric strength for the backs.

TABLE 1. Characteristics of all players and per playing position. Data reported as mean values with SD or percentages

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TABLE 2. *Isometric endurance and peak isometric cervical strength scores. All values are reported in seconds (s), Newtons (N) or Newtons per kilogram body weight (N/kg)*

TABLE 3: *Multiple linear regression analysis estimating the change in peak isometric force and endurance, adjusted for playing position. Values reported in Newton (N) and seconds (s). *,p<0.01, df, degrees of freedom*

There was a statistically significant association between peak flexion strength and flexion endurance: for every one second increase in endurance, an increase of 0.79N (0.29 to 1.29) was observed in isometric strength (p=0.003). The scatter plots in Supplementary FIGUREs 5 and 6 demonstrate the difference between positions. There was no statistically significant relationship between extension endurance and peak extension isometric strength (p=0.64).

There was a positive association between age and peak isometric strength in lateral flexion (p=0.02 for right and p=0.01 for left lateral flexion) for all position groups (Supplementary Table 1, Supplementary FIGURE 7.). For every year older, right lateral flexion peak force increased by 4.16N and left lateral flexion increased by 5.17N. There were no significant associations between age and peak isometric strength in either flexion or extension (Supplementary TABLE 1), or isometric endurance (flexion p= 0.58, extension p=0.99). Body weight was strongly associated with peak isometric strength in all four directions, across all playing positions (Supplementary TABLE 2, Supplementary FIGURE 8). For every 1kg increase in body weight an increase of 2.5N in strength for flexion, 4.6N for extension, 3.08N for right lateral flexion and 3.7N for left lateral flexion was observed (Supplementary TABLE 2). There were no associations between body weight and cervical muscle endurance.

Front row forwards were stronger than backs in all four peak isometric strength tests when adjusting for player height (Supplementary TABLE 3). There was a negative association between height and extension endurance, which was similar across all groups, where every 1cm increase in height resulted in a 0.75 second decrease in extension endurance time (p=0.02) (Supplementary TABLE 3, Supplementary FIGURE 9).

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4. DISCUSSION

Position-specific normative data are presented for isometric cervical muscle strength and cervical muscle endurance in professional male rugby players. Peak isometric strength was associated with age and body weight. Front row forwards were stronger and had greater endurance than other forwards and backs. There was a significant relationship between

isometric endurance and peak isometric strength in flexion, with substantial variability among players and position when considering age, height and body weight.

Position

Generally, players are categorised as either 'forwards' (positions 1–8) or 'backs' (positions 9–15), with the strength profile, body composition, physical characteristics and positional demands differing between these groups.[26] We subdivided the forwards group and observed large ranges in both the flexion and extension endurance. This highlights the importance of suitably categorising position groups to appropriately assess players, including both force and endurance.[2,16,24,28]

Cervical muscle strength

There were statistically significant differences in peak isometric cervical strength between each position group in all four directions, with the front row forwards producing the highest values. A study on isokinetic neck strength of South African rugby players found that second rows (other forwards) produced greater force during peak isometric flexion, and both second and back rows produced greater force on side flexion (lateral movements).[25] The peak isometric strength was comparable to our study for the forward/flexion measure only.[30] The backward/extension measure at 230N for backs and 328N for forwards was notably smaller than that observed in our study. A prone position, pushing against gravity may have had some impact on the isometric extension strength and the differing methodology must be considered when comparing to our data. In contrast, Geary et al[21] tested a younger cohort of rugby players using a handheld dynamometer and report higher peak isometric strength scores in all directions. Specific cut-offs may be difficult to identify, and strengthening of the cervical muscles in all directions of movement is likely clinically-relevant.

When the raw strength scores were adjusted for bodyweight, the statistical significance across the position groups were maintained. Although normalizing to bodyweight may be valuable in some cases, the raw strength scores may hold a true indication of the differences between groups.

Height and Weight

Height was negatively associated with extension endurance, with taller players demonstrating lower endurance scores, however the effect was small (p= 0.02). Although taller players may have smaller cervical cross-sectional muscle, this remains unsubstantiated.

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Front row forwards held similar weights to the other forwards, as this was normalised to body weight as previously described, but body composition and distribution of muscle bulk vary widely between these groups. Also, the positional demands of front row forwards may influence their extension endurance results as they are required to sustain isometric holds into extension for scrum stability.[7] Other forwards do not have the same physical demand, although they are also in extended neck positions during scrummaging.[30]

There was a small correlation between isometric endurance and peak isometric flexion strength (p=0.003), but no significant relationship between isometric endurance and peak isometric strength in extension. Both tests may potentially have a role to play in risk factor identification and concussion prevention.[42]The sustained cervical muscle demands in rugby make endurance capacity a key focus for physical preparation. Forwards engage in an average 19 scrums per match[45] with over 200 tackle events observed across all position groups in a professional match.[12] Using the adapted cervical endurance test described, when the weight held was adjusted to body weight, there were no significant differences between position groups for extension or flexion. Other forwards had an average of 50.2 seconds for extension compared to 57.1 and 58.2 for the front row forwards and backs respectively, and while this was not statistically significant, the 14% difference may be relevant in the clinical setting. Slight improvements in muscle endurance may provide players with greater resistance against fatigue and mitigate their risk of injury.

Cervical muscle strength that meets the demands of the position is a vital element of injury prevention.[13] As catastrophic events have substantially reduced with law changes to the game and greater awareness around safety,[27] the focus is now on reducing significant time loss injuries. Previous research investigated scrum-related cervical spine injuries in front row forwards, however there is a lack of similar research in other forwards and backs.[38] Although backs do not participate in the scrum, they are more frequently removed for a head injury assessment in professional rugby.[35,37,46] The normative data presented here will assist practitioners to develop appropriate standards for playing position related to preventing spinal injuries by addressing cervical strength.

Age

Cervical strength was positively associated with age, but this was only significant for right and left lateral flexion with a small effect size (p=0.01). Strength may gradually increase as players get older. When adjusting for age, significant differences between the position groups' peak isometric strength scores indicated that age was not a strong predictor of peak strength, despite

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312 the positive relationship. Age may represent a proxy of exposure to rugby, which may lead to
313 additional strength gains. Mean age in our study was 25.3±3.4 years, suggesting a more
314 homogenous group and it may have had insufficient power to identify age-related differences.

315 Previous research highlighted significant differences between age grade (under 18 and under 20
316 years old) and senior front row forwards' peak isometric strength results, where the mean age
317 of the age-grade players was 19±1 compared to the seniors s 27±5 years.[10,14]

318 While age is one factor to consider, senior players were also significantly heavier. The results
319 presented here, suggest that weight may be the greatest predictor of cervical strength. This
320 relationship was true across all position groups. The strongest effect size was seen for extension,
321 where a 1kg increase in body weight led to a 4.6N increase in isometric strength. The heavier
322 front row forwards were also the strongest. However, it is important to note the specific
323 positional demands on a player and whether increased body weight would negatively impact
324 other elements of performance.

325 In non-athletic populations, peak isometric strength measured by fixed frame dynamometry
326 varied by 300N in extension and 180N in flexion.[9] Similarly, large variability within the current
327 cohort was identified, despite sub-dividing into categories more reflective of body type and
328 position. This variability should be considered when assessing cervical strength in a practical
329 setting. Clinicians must interpret mean and normal distribution of data within their cohort, as
330 well as individual characteristics around height, body weight, age, and position when
331 interpreting data against these and other normative ranges.

332 Limitations

333 Testing was conducted in a population of elite male rugby players, and we urge care when
334 generalising these findings to other levels of competition, sex, or sports. Although we performed
335 a power calculation that suggested our study was adequately powered, a sample size of n=136
336 may not denote adequate representation.[3] Testing took place at one time-point during pre-
337 season training, however the type and volume of training completed in the weeks prior to testing
338 differed across the cohort. Therefore, results may not reflect players' physical state during the
339 whole season. While efforts were made to conduct all testing on the same day, this was not
340 feasible in all cases due to the schedules of professional athletes. We excluded players who had
341 undergone cervical surgery in the three months prior to the testing, but other surgeries may also
342 have influenced the results. Repeated measures throughout the season may have provided a
343 more accurate representation of cervical muscle strength due to the quality and quantity of
344 training and occurrence of injury in-season.

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Although good to excellent reliability has been established for the isometric strength test, criterion-related validity has not. The highest of the three repetitions was recorded, which does not account for potential variability between the measurements. Pragmatically, we tested extension before flexion with a five minute rest period, although we cannot rule out bias and the potential influence of fatigue on the results. Another consideration is visual assessment of deviation from test position, which resulted in termination of the peak endurance test.

5. Conclusion

This cohort study determined position-specific cervical muscle peak strength and endurance in a cohort of elite men's rugby players. There was a significant relationship between isometric endurance and peak isometric cervical strength in flexion for this cohort of elite men's rugby players. Front row forwards are stronger and have greater endurance. Together with age and body weight, players presented with substantial intraindividual variability, which should be considered when interpreting cervical strength and endurance results.

Key Points

Findings:

- Front row forwards had significantly stronger cervical muscles than other forwards and backs. Strength programmes in rugby players should reflect the differing strength profiles of each playing position and the corresponding demands of that position.

Implication:

- There was a significant relationship between peak isometric flexion strength and flexion endurance. However, no such relationship exists between peak isometric extension strength and endurance. When assessing cervical muscle strength, these results suggest appreciation for intraindividual variability, considering both strength and endurance.

Caution:

- The large variation in strength scores among the playing group demonstrates the importance of stratify by playing position when evaluating strength scores in rugby players.

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Tables:

Table 1. Characteristics of all players and per playing position. Data reported as mean values with SD.

	All n=136	**Backs** n=61	**Front row forwards** n=37	**Other forwards** n=38	**p-value (ANOVA)**
Age (years)	25.3 ± 3.4	24.3 ± 3	26 ± 3.1	26.2 ± 3.8	0.006
Height (cm)	185.9 ± 7	182.3 ± 5.5	183.5 ± 3.4	193.8 ± 5.2	<0.001
Weight (kg)	103 ± 13.2	91.5 ± 7.7	113.9 ± 10	110.7 ± 6.4	<0.001

Table 2. Isometric endurance and peak isometric cervical strength scores. All values are reported in seconds (s), Newtons (N) or Newtons per kilogram (N/kg)

Peak isometric strength (N)

	All n=129	**Backs n=58**	**Front row forwards n=35**	**Other forwards n=36**	**p-value (ANOVA)**	
Flexion	275 ± 65	247 ± 48	319 ± 70	276 ± 61	<0.001	0.641
BW adjusted	2.53 ± 0.79	2.57 ± 0.76	2.63 ± 0.82	2.37 ± 0.78	0.07	0.027
Extension	429 ± 104	380 ± 82	516 ± 116	424 ± 62	<0.001	0.281
BW adjusted	3.95 ± 1.22	3.94 ± 1.19	4.25 ± 1.37	3.64 ± 1.05	0.02	0.079
Left lateral flexion	306 ± 73	282 ± 68	348 ± 70	304 ± 66	<0.001	0.125
BW adjusted	2.82 ± 0.90	2.94 ± 0.94	2.86 ± 0.87	2.60 ± 0.83	0.04	0.035
Right lateral flexion	308 ±70	277 ± 65	362 ± 59	307 ± 53	<0.001	0.245
BW adjusted	2.84 ± 0.87	2.88 ± 0.93	2.99 ± 0.86	2.63 ± 0.75	0.01	0.052

Isometric endurance (s)

	All n=136	**Backs n=61**	**Front row forwards n=37**	**Other forwards n=38**	**p-value (ANOVA)**	**Effect size (ω^2)**
Extension	55.7 ± 17.1	58.2 ± 16.9	57.1 ± 19.9	50.2 ± 13.4	0.06	0.03
Flexion	52.9 ± 20.1	52.5 ± 18.9	54.7 ± 22.9	51.8 ± 19.7	0.81	0.01

BW: body weight, ω^2, Cohen's omega squared

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Table 3: Multiple linear regression analysis estimating the change in peak isometric force and endurance, adjusted for playing position. Values reported in Newton (N) and seconds (s). *,p<0.01, df, degrees of freedom

	Peak isometric strength			
	Flexion	**p-value**	**Extension**	**p-value**
Backs (comparator)	247.8 (233.2 to 262.3)		379.7 (356.9 to 402.5)	
Peak isometric strength				
Front row forwards	70.9 (47.2 to 94.6)	<0.001	136.9 (99.8 to 174)	<0.001
Other forwards	29.9 (6.4 to 53.4)	0.02	45.3 (8 to 82.6)	0.02
Isometric endurance strength				
Flexion (s)	0.79 (0.29 to 1.29)	0.003		
Extension (s)			0.23 (-0.72 to 1.18)	0.64
R^2	0.27		0.30	
Adjusted R^2	0.25		0.28	
Residual standard error (df = 125)	56.6		88.37	
F-statistic (df = 3; 125)	15.07*		17.51*	

FIGURES

FIGURE 1 *Extension endurance test with head harness***.**

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FIGURE 2 *Flexion endurance test with head harness*

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